

COMSATS University Islamabad

Sahiwal Campus

(Department of Computer Science)

INFORMATION SECURITY

Lab Assignment # 3

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Q #1:

Conduct research on the latest attacks targeting key exchange algorithms (excluding MitM attacks) and develop an implementation in Python). Also explain your research in one paragraph.

Cache Timing Attack on RSA Key Exchange:

Cache timing attacks are a category of **side-channel attacks** that exploit variations in a system's response time to infer sensitive data such as private keys. These attacks are particularly potent against cryptographic algorithms like **RSA**, especially during modular exponentiation, where the time to process a '1' bit may differ from a '0' bit due to branch conditions or memory access patterns. Unlike Man-in-the-Middle (MitM) attacks, cache timing attacks do not require active interception of data. Instead, they exploit local or co-located system access, such as in **cloud computing environments** where attackers may share physical hardware with the target. This threat becomes even more critical when running cryptographic processes on shared CPUs or virtual machines. Notable historical cases include timing leaks in OpenSSL implementations. To combat these risks, modern cryptographic libraries employ **constant-time execution**, **cache isolation**, and hardware-level protections.

Python Implementation: Cache Timing Side-Channel Simulation

```
import time
import random

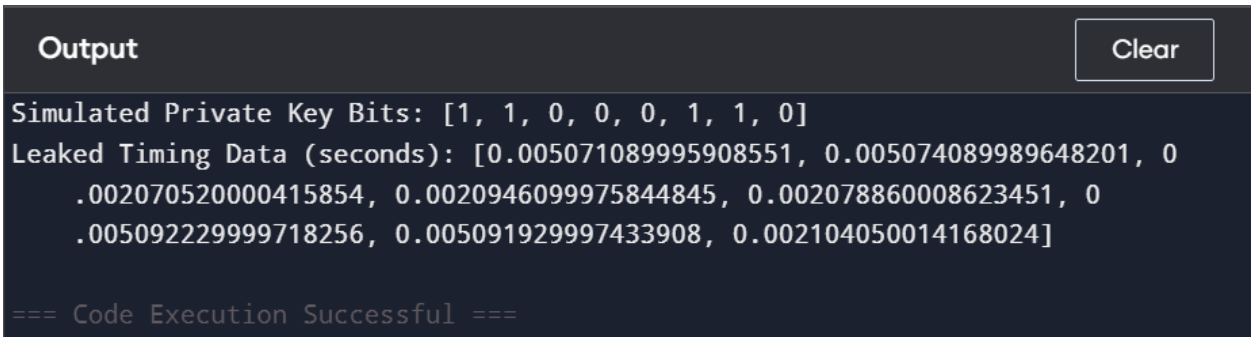
private_key = [random.choice([0, 1]) for _ in range(8)]

def modular_exponentiation(bit):
    start = time.perf_counter()
    if bit == 1:
        time.sleep(0.005)
```

```
    else:
        time.sleep(0.002)
    end = time.perf_counter()
    return end - start

timings = []
for bit in private_key:
    duration = modular_exponentiation(bit)
    timings.append(duration)

print("Simulated Private Key Bits:", private_key)
print("Leaked Timing Data (seconds):", timings)
```



Explanation and Mitigations

This simulation models a scenario where an attacker can deduce bits of an RSA private key by observing the timing differences in modular exponentiation. Each bit's processing time mimics the behavior of a vulnerable RSA implementation, where different code paths introduce measurable delays. Although simplified, it illustrates how timing information leaks can serve as a covert channel for key recovery.

Mitigations:

- **Constant-Time Algorithms**: Always execute cryptographic operations in the same number of cycles, regardless of input.
- **CPU Cache Partitioning**: Prevent attackers from observing shared cache behavior.
- **Physical & VM Isolation**: Use isolated environments in cloud systems.
- **Compiler & Library-Level Protections**: Libraries like OpenSSL and libsodium now implement side-channel-resistant methods by default.

References

- Kocher, P. C. (1996). *Timing Attacks on Implementations of Diffie-Hellman, RSA, DSS, and Other Systems.*
- Bernstein, D. J. (2005). *Cache-timing attacks on AES.*
- Wikipedia - Side-channel attack
- OpenSSL Developer Docs: https://www.openssl.org/docs